HOMEWORK 8.6 · UNIT 8 · NC.8.G.4 · NC.8.G.5 · NC.8.G.3

Dilations Day 4

Targets: G.4-7, G.4-2, G.4-3, G.4-4, G.5-3, G.4-9, G.3-5, G.4-8

Name _____ Date _____ Block _____

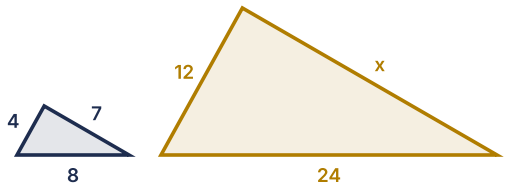
Remember: In similar figures, write a **proportion** with matching sides: $\frac{\text{small}}{\text{large}} = \frac{\text{small}}{\text{large}}$. Or use the scale factor.

Example: $\frac{4}{6} = \frac{10}{x} \rightarrow 4x = 60 \rightarrow x = 15$.

SCORE 3.0 Approaching · the basics

Targets: G.4-7, G.4-2, G.4-3, G.4-4, G.5-3

1 The triangles are similar. Find x. G.4-7



2 Solve: $\frac{3}{4} = \frac{x}{20}$ G.4-7

(A) $x = 15$ (B) $x = 19$ (C) $x = 60$ (D) $x = 5$

3 A triangle with sides 10, 12, 14 is similar to one with sides 15, 18, 21. Find the scale factor from small to large. REVIEW 8.5 G.4-2

4 Are these ratios equal? $\frac{4}{10}$ and $\frac{6}{14}$ G.4-3

Yes No

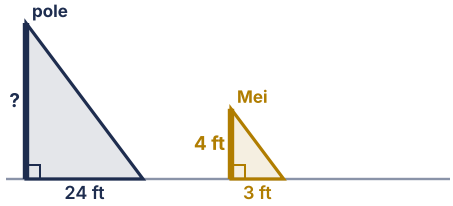
Why? _____

5 △ABC ~ △DEF. m∠A = 35° and m∠B = 90°. Find m∠F. REVIEW 8.5 G.4-4,G.5-3

SCORE 4.0 Proficient · on target

Targets: G.4-9, G.4-7, G.3-5

6 Mei is 4 ft tall and casts a 3-ft shadow. A pole casts a 24-ft shadow at the same time. How tall is the pole? G.4-9



7 **Sam** made a mistake. Circle it. Explain it. Fix it. G.4-7

SAM'S WORK

Similar figures: small sides 3 and 7, large sides 9 and x.

$9 - 3 = 6$, so $x = 7 + 6 = 13$

8 A(4, −8) is dilated about the origin to A′(1, −2). Find the scale factor. REVIEW 8.4 G.3-5

Ⓐ $\frac{1}{4}$ Ⓑ 4 Ⓒ −3 Ⓓ $\frac{1}{2}$

SCORE 5.0 Mastery · go deeper

Targets: G.4-9, G.4-8

9 Ana is 5 ft tall and casts a 2-ft shadow. A tree casts a 14-ft shadow at the same time. How tall is the tree? Explain why your method works. G.4-9

The tree is _____ . This works because _____ .

10 A 2 by 5 rectangle and a 6 by 15 rectangle. Are they similar? Explain using a dilation. G.4-8

They _____ similar. A dilation by _____ maps _____ .